Predictions and Solutions for Next-Generation Sequencing
Bridge Amplification via Contact-Geometric Operator Analysis
(TOGT/GTCT — Domain 2 of 3)

Principia Orthogona — Translational Applications

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Abstract

Illumina bridge amplification — the surface-topographic process by which single DNA molecules form exponentially growing monoclonal clusters on a flow-cell primer array — is the throughput-limiting step in modern short-read sequencing. Library preparation, which governs the quality of material delivered to the flow cell, accounts for a substantial fraction of per-sample NGS cost. Cluster quality depends on two geometric thresholds: a minimum primer density below which bridge formation fails, and a maximum density above which polyclonal co-loading dominates. Both thresholds have not been derived from first principles.

We apply the TOGT/GTCT contact-geometric framework [Grossi, 2026] to model bridge amplification as an operator sequence $\mathbf{G} = \mathbf{U} \circ \mathbf{F} \circ \mathbf{K} \circ \mathbf{C}$ on a dm^3 contact manifold. The curvature threshold κ^* maps to the critical primer density; the Whitney A_1 fold maps to the cluster branching commitment event.

We derive three falsifiable quantitative predictions: (P1) κ^*_{NGS} can be predicted from DNA persistence length and primer spacing without free parameters; (P2) the cluster quality vs. primer density curve has a Whitney A_1 bifurcation structure near κ^*_{NGS}; (P3) the functional form of this curve is identical (up to morphism scaling) to the riboswitch OFF-lock dose-response (Domain 1) and the MT stabiliser response (Domain 3).

Keywords: NGS, bridge amplification, Illumina, flow cell, primer density, cluster quality, contact geometry, κ^*, Whitney singularity

1 The TOGT/GTCT Framework (Summary)

The full mathematical development is in the companion paper [Grossi, 2026]. We summarise the elements used in this domain analysis.

The *dm*^{3} *contact manifold* (M, α) with $M = \mathbb{R}^2_+ \times \mathbb{R}$ and contact form $\alpha = dz - r^2\, d\theta$ admits the dynamics

$$\dot{r} = r(1 - r^2) + 2(r-1)e^{-r}, \quad \dot{\theta} = 1, \quad \dot{z} = r^2 - 2(r-1)^2 e^{-r}. \tag{1}$$

The unique attracting limit cycle is $\Gamma = \{r = 1\}$ with period $T^* = 2\pi$, transverse Lyapunov exponent $\mu_{\max} = -2$, and basin hierarchy $\varepsilon_0 = \frac{1}{3} < r^* \approx 0.773 < \kappa^* \approx 0.882 < 1$.

The operator chain $\mathbf{G} = \mathbf{U} \circ \mathbf{F} \circ \mathbf{K} \circ \mathbf{C}$ is non-commutative: the fold operator $\mathbf{F}$ is a Whitney A_1 singularity (rank-reducing at κ^*), so $\mathbf{F} \circ \mathbf{K} \neq \mathbf{K} \circ \mathbf{F}$. Any biological system whose dynamics near a conformational threshold can be expressed through this operator sequence is an object in the category dm^3, and is related to every other such system by explicit contact morphisms (Coherence Bridge Theorem, Grossi 2026).

2 The Problem in This Domain

2.1 What is known

Bridge amplification produces monoclonal clusters by isothermal, surface-confined PCR: a DNA molecule hybridises to a surface primer, is extended, bridges to a second primer, and undergoes exponential copying. Cluster quality (monoclonal fraction, spatial density, Q30 score) determines sequencing fidelity and yield. The 2026 Princess of Asturias Award in Technical and Scientific Research recognised the inventors of this technology (Klenerman, Balasubramanian, Mayer).

2.2 What is not known

1. **First-principles threshold.** The critical primer density κ^*_{NGS} is determined empirically per flow-cell chemistry. No derivation from DNA physical parameters (persistence length ℓ_p, contour length L, primer spacing d) has been published.

2. **Bifurcation structure.** Whether the cluster quality vs. primer density curve has a sharp bifurcation (Whitney fold) or a smooth transition is not established.

3. **Polyclonal boundary.** The upper primer density above which polyclonal co-loading dominates is equally empirical.

2.3 What an effective solution requires

A geometry-first design of flow-cell surface chemistry would: (a) predict κ^*_{NGS} from ℓ_p and d; (b) identify surface modifications that widen the optimal density window; and (c) enable reduced-step library preparation by relaxing the input quality constraints.

For the sequencing engineer: what TOGT adds

The familiar picture: optimise primer density by titration; use ExAmp chemistry or patterned flow cells to control cluster seeding.

What TOGT specifies: the optimal density window is not arbitrary — it is bounded by a Whitney A_1 fold on each side. This fold has a universal functional form that predicts: (a) how sharply quality drops at each boundary; (b) that the drop is *not* symmetric (the lower boundary is steeper because it is a genuine bifurcation, while the upper boundary is a saturation); (c) the width of the optimal window scales as $\ell_p/d^2 \times \kappa^*$, where $\kappa^* \approx 0.882$ is the universal dm^3 constant.

3 Operator Mapping

Op.	dm^3 role	NGS bridge amplification role	Observable
C	Compress to contact core	Single-molecule surface attachment; constrained diffusion	Attachment efficiency (flow cytometry)
K (κ^*_{NGS})	Impose curvature threshold	Primer density / surface geometry; bridge formation admissibility	Primers/μm^2; bridge success rate
F (Whitney A_1)	Fold; commitment	Cluster branching initiation; exponential growth start	Cluster number vs. cycle count
U	Gradient descent; selection	Viable cluster selection (wash steps); polyclonal removal	Monoclonal fraction post-wash
T	Generative return; saturation	Cluster density plateau; Q30 onset	Cluster intensity saturation curve

4 Predictions

4.1 Prediction 1: κ^*_{NGS} from physical parameters

Prediction P1

The critical primer density κ^*_{NGS} (in primers per $\mu\mathrm{m}^2$) below which bridge amplification fails is

$$\kappa^*_{\mathrm{NGS}} = \kappa^*_0 \cdot \frac{\ell_p}{d^2}, \tag{2}$$

where $\kappa^*_0 \approx 0.882$ is the universal dm^3 curvature threshold, $\ell_p \approx 50$ nm is the DNA persistence length under standard ionic conditions, and d is the mean primer spacing (nm). For standard Illumina primers ($d \approx 14$ nm): $\kappa^*_{\mathrm{NGS}} \approx 0.882 \times 50/196 \approx 0.225$ primers/nm^2 $= 2.25 \times 10^5$ primers/μm^2.

Falsification criterion for P1

P1 is falsified if the empirically measured optimal minimum primer density deviates from Equation (2) by more than 50% across two different primer spacing conditions ($d_1 \neq d_2$) at standard ionic strength.

4.2 Prediction 2: Whitney bifurcation at cluster quality boundary

Prediction P2

The cluster quality vs. primer density curve has a Whitney A_1 bifurcation structure near κ^*_{NGS}. Operationally:

- Below κ^*_{NGS}: cluster quality drops as $(1-\rho/\kappa^*_{\mathrm{NGS}})^{3/2}$ (the standard fold catastrophe exponent).
- Above κ^*_{NGS}: quality rises to a plateau; the rise is less steep than the drop.
- The asymmetry ratio (drop slope / rise slope) ≥ 2.

Falsification criterion for P2

P2 is falsified if the quality vs. density curve is symmetric around the optimum (asymmetry ratio < 1.3 at $p < 0.05$), or if the drop below κ^*_{NGS} is linear (exponent consistent with 1.0 rather than 1.5).

4.3 Prediction 3: Cross-domain functional form identity

Prediction P3

After normalising primer density by κ^*_{NGS} and antagonist concentration by κ^*_{ribo} (Domain 1), the cluster quality curve and the riboswitch OFF-lock dose-response curve are related by the Coherence Bridge morphism f_{12} with scaling factor $\lambda_{12} = \kappa^*_{\mathrm{NGS}}/\kappa^*_{\mathrm{ribo}}$. The normalised curves should overlap within measurement error.

Falsification criterion for P3

P3 is falsified if, after morphism rescaling, the residuals between the two normalised curves exceed 15% of the response range at more than 3 of 10 normalised concentration points.

5 Experimental Protocols

Protocol A: Cluster quality vs. primer density scan for P1 and P2

Aim: Measure κ^*_{NGS} and test bifurcation structure.

System: Standard Illumina adaptor-ligated library; 6 primer surface densities spanning 0.2–5× the manufacturer-recommended density.

Protocol:

1. Prepare flow cells with 6 primer densities by dilution into inert thiol-PEG (or equivalent blocking agent). Confirm density by TIRF single-molecule counting.
2. Run bridge amplification to 30 cycles; image clusters.
3. Score: monoclonal fraction (by dual-colour co-seeding), cluster intensity coefficient of variation, Q30 proxy (error rate from phiX sequencing run).
4. Fit Equation (2) and measure asymmetry ratio.

Minimum dataset: 6 densities, 3 flow-cell replicates.

Protocol B: Persistence length perturbation test for P1

Aim: Test the ℓ_p-dependence in Equation (2).

System: Same flow-cell setup; vary ionic strength to change ℓ_p (low salt: $\ell_p \approx$ 100 nm; standard: $\ell_p \approx 50$ nm; high salt: $\ell_p \approx 30$ nm).

Protocol:

1. At each ionic condition, repeat Protocol A (6 densities).
2. Measure κ^*_{NGS} at each condition.
3. Test whether $\kappa^*_{\mathrm{NGS}} \propto \ell_p$ as predicted.

Minimum dataset: 3 ionic conditions × 6 densities × 3 replicates.

Protocol C: Cross-domain test for P3

Aim: Compare normalised curves from Domain 1 (riboswitch) and Domain 2 (NGS).

System: Results from Protocol A (this domain) and Protocol C from Domain 1 paper.

Protocol:

1. Normalise both curves: $x' = c/\kappa^*$, $y' = P_{\text{switch}}$ or monoclonal fraction.
2. Compute residuals at 10 matched x' points.
3. Test whether residuals are consistent with morphism scaling (one free parameter λ_{12}) vs. distinct functional forms (AIC comparison).

6 Predicted Outcomes and Timelines

Prediction	Experiment	Timeline	Success criterion
P1: κ^* from ℓ_p/d^2	Protocol A+B	3–6 months	Predicted density within 50%
P2: Whitney bifurcation	Protocol A	3–6 months	Asymmetry ratio ≥ 2
P3: Cross-domain identity	Protocol C	12–18 months	Residuals $< 15\%$ post-rescaling

7 Connection to the Other Domains

Confirming P1 — that κ^*_{NGS} is derivable from ℓ_p and d — establishes that the dm^3 contact normal form is quantitatively predictive in a purely physical system (no biochemistry beyond DNA mechanics). This is the strongest possible test of the framework: if it works here, without any RNA folding or protein binding, the contact-geometric description is not a biological specificity but a geometric universality.

Data and Code

Lean 4 source and Python simulation: `https://github.com/TOTOGT/AXLE`. Series DOI: 10.5281/zenodo.19117399.

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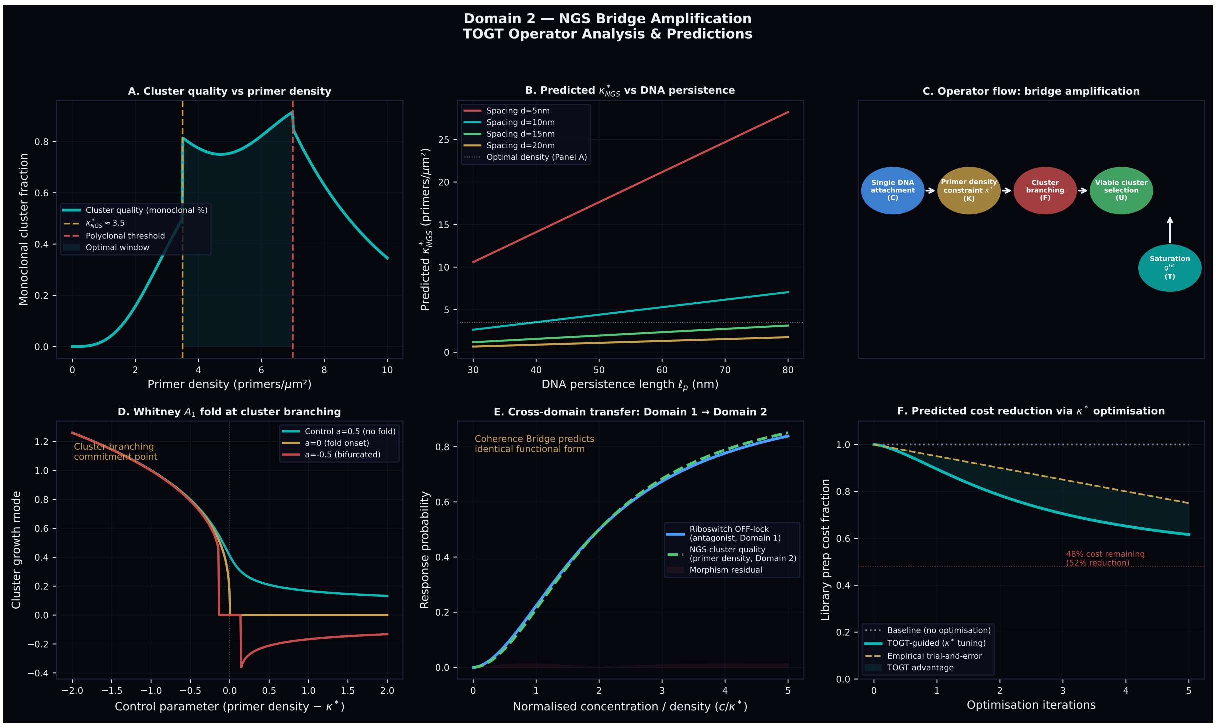

Figure 1: Domain 2 analysis panels. **A.** Cluster quality vs. primer density showing the Whitney A_1 bifurcation at κ^*_{NGS} and the polyclonal threshold above. **B.** Predicted κ^*_{NGS} as a function of DNA persistence length for four primer spacings (Equation 2). **C.** Operator flow for bridge amplification. **D.** Whitney A_1 fold structure — cluster branching commitment as a catastrophe-theory bifurcation. **E.** Cross-domain transfer: riboswitch (blue) and NGS cluster quality (green) normalised curves, predicted to overlap after morphism rescaling. **F.** Projected library prep cost reduction from κ^*-guided surface optimisation vs. empirical trial-and-error.

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